Conditional Groups

Say you were validating a list of banquet attendees:

```
"Mr Jones and Sally",
"Mr Bond and Ms Moneypenny",
"Samson and Delilah",
"Dr Jekyll and himself",
"Ms Hinky Smith and Ms Jones",
"Dr Wood and Mrs Wood",
"Thelma and Louise"
```

The rule is that if the first person in the list has a title, then so should the second. This means that the first and fourth lines in this list are invalid.

We can start with a pattern to match a line with an optional title and a name. We know we've reached the end of the name when we find the word *and* with spaces around it. Since we're using the x modifier the regex engine will ignore whitespace to allow us to change the layout of the regex, we need to explicitly identify the spaces that we are matching.

```
re = %r{ (?:(Mrs | Mr | Ms | Dr )\s)? (.*?) \s and \s }x
"Mr Bond and Ms Monneypenny".match(re).captures # => ["Mr", "Bond"]
"Samson and Delilah".match(re).captures         # => [nil, "Samson"]
```

Let's try again. We've defined the regexp with the x (extended) option so we can include whitespace. We also used the ?: modifier on the group that defines the optional title followed by a space. This stops that group getting captured into $1. We do however use a nested group to capture just the title part.

So now we need to match the second name. We can start with the same code as for the first:

```
re = %r{
  (?:(Mrs | Mr | Ms | Dr )\s)? (.*?)
  \s and \s
  (?:(Mrs | Mr | Ms | Dr )\s)? (.+)
}x
"Mr Bond and Ms Monneypenny".match(re).captures # => ["Mr", "Bond", "Ms",
                                                # .. "Monneypenny"]
"Samson and Delilah".match(re).captures         # => [nil, "Samson", nil,
                                                # .. "Delilah"]
```

Before we go any further, let's clean up the duplication using a named group:

tut_regexp/attendee_validator_1.rb

```
re = %r{
  (?:(?<title>Mrs | Mr | Ms | Dr )\s)? (.*?)
  \s and \s
  (\g<title>\s)? (.+)
}x
match_data = re.match("Mr Bond and Ms Monneypenny")
match_data[0]         # => "Mr Bond and Ms Monneypenny"
match_data[:title]    # => "Ms"

second_match = re.match("Samson and Delilah")
second_match[0]       # => "Samson and Delilah"
second_match[:title]  # => nil
```